CHAPTER 7

Primitives and differential equations

S52, S53, S55, S58

Before you start · Derivation and models

Without a calculator, in six minutes. Respond briefly. Mobilize derivative, exponential and units. Then choose the targeted pick-up indicated under the questions.

1. Derivating $F(x) = x^2$.

2. Find an affine function whose derivative is 3.

3. Derive 3 e^{2x}.

4. In $f(t) = 80e^{-0,1t}$, calculate $f(0)$.

5. Convert 90 min to hours.

6. What is worth e^0 ?

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Retake the derivative x squared. **2.** Write mx+b and drift to determine m. **3.** Identify the factor 2 from the composition; R36. **4.** Substitute t=0 before calculating the exponential. **5.** Divide minutes by 60; R67. **6.** Retrieve the value of the original exp. Questions R serve as support for review with the teacher, even if their collective ritual is planned later.

Objectives

Calculate antiderivatives; solve a differential equation with initial condition; interpret an approximation of Euler.

Course notes

Primitives over an interval

It is admitted at this stage that any function continues over an interval admits antiderivatives; the connection with Chapter 8 will be prepared for this purpose. f over an interval I is a detritable function F such that $F' = f$ on I. If F is a antiderivative, all antiderivatives on I are $F + C, C \in \mathbb{R}$.

Demonstration. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a disjunct domain, the constants may differ according to the intervals.

Table of antiderivatives · inverse reading of derivatives

$f(x)$	$F(x)$
x^n $(n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ over an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\ \overline{x}\quad (x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be reinvested
After the study of trigonometry.

Recognition of membership

For a function u Drifting, antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on intervals where expressions have a meaning. Linearity allows to process sums.

Example. A antiderivative of 6 $x(x^2+1)^2$ is $(x^2+1)^3$; a antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation connects an unknown function to its derivatives.
over a specified interval.

Common solutions

$y' = ay$ a for solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ a for solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a particular known solution of $y' = ay + f(x)$, solutions are $y = g + Ce^{ax}$.

Demonstration for $y' = ay$**.** If y is a solution, $z(x) = e^{-ax}y(x)$ checks $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtract the constant solution $-\frac{b}{a}$ the same equation. For a second variable member, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$then $C = -2$. Solution 3 $-2e^{-2x}$ growth to balance 3 when $x \to +\infty$.

Method of production

For $y' = ay + b$, with one step $h > 0$, we're approaching $y(t_k)$ by Y_k by $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. It is an approximation using tangent, not equality on the exact solution.
A smaller step can improve approximation; numerically control error on a solution
known.

Dive backwards and model an evolution

7.1 Recognize and control a antiderivative

First use: S52.

Definition · Primitive over an interval

A F function is a antiderivative f on I if F is driftable on I and $F' = f$ I. Interval is part of the data. Finding a antiderivative is an inverse calculation; the derivation allows to Check it out.

Example Solved · Correct a Factor

For $f(x) = 3e^{2x}$, $F(x) = \frac{3}{2}e^{2x}$ is appropriate because $F'(x) = \frac{3}{2} \times 2e^{2x} = 3e^{2x}$. For $g(x) = 2x/(1+x^2)$ on $\mathbb{R}$, the antiderivative $G(x) = \ln(1+x^2)$ uses a positive argument. All antiderivatives obtain by adding a constant.

A1 · Immediate application Determine a antiderivative of 4 $x^3 - 2x + 1$, of e^{-3x} and 1 $/(2x+1)$ on a suitable interval. Check by bypass.

7.2. A constant, a vertical translation

First use: S52.

Demonstration. If $F' = G' = f$ over interval I, then $(F-G)' = 0$ on I. A derivative function zero over an interval is constant; therefore $F - G = C$. On a domain consisting of several intervals Disjointed, the constant can be different over each interval. □

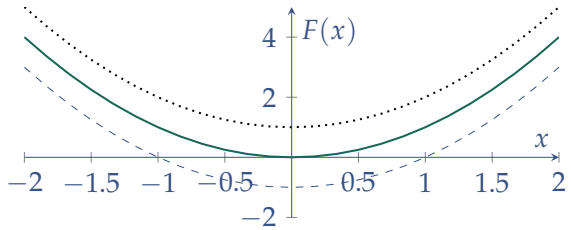


Figure 7.1. Three antiderivatives of $2x$: On the same apses, the tangents have the same slopes.

Solved Example · A value sets the translation

The antiderivatives of $2x$ are $x^2 + C$. Imposing F(0)=1 selects $F(x) = x^2 - 1$. The graph is Translating a unit downwards without changing the slopes.

A2 · Partial guidance Determine the antiderivative of $2x$ which is worth 3 in 1. Then explain why all the antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$ are not described by a single constant common to both Half straight.

7.3. Unknown is a function

First use: S55.

Ownership · Equation $y' = ay$

For $a \in \mathbb{R}$, solutions on $\mathbb{R}$ are $y(t) = Ce^{at}$, $C \in \mathbb{R}$. An initial value $y(0) = y_0$ imposes $C = y_0$.

Demonstration. A function of this form is well solution. Conversely, if it is solution, $z(t) = y(t)e^{-at}$ a for derivative $(y' - ay)e^{-at} = 0$, so is constant over the interval. $y(t) = Ce^{at}$. The evidence checks the solutions and shows that none is missing. □

Example resolved · A condition in another instant

For $y' = 3y$ and y(1)=2, $y(t) = Ce^{3t}$then $Ce^3 = 2$. Therefore $C = 2e^{-3}$ and $y(t) = 2e^{3(t-1)}$. The constant is equal to the value imposed only if it is given in 0.

A3 · Immediate application Solve $y' = -2y$ with $y(0) = 5$. Is the number 0 a solution possible differential equation alone ? Of this problem with initial condition ?

7.4 Subtract a known solution

First use: S55 and S58.

Example resolved · Second variable member

For $y' = -y + 2t + 1$, seek a particular solution $p(t) = at + b$ led to $a = -at - b + 2t + 1$. Identify the coefficients given $a = 2$, $b = -1$. If y is a solution, $z = y - p$ checks $z' = -z$. With $y(0) = 3$, we get $z(0) = 4$, so $y(t) = 2t - 1 + 4e^{-t}$.

A4 · Partial guidance Solve $y' = 2y - 4$ with $y(0) = 3$ by subtracting a constant solution. Check the formula and initial value.

A5 · Repair Evidence One student checked that a function satisfies a differential equation, and then "I have found all the solutions." Why is this conclusion insufficient? Illustrate with $y' = y$.

7.5 Discrete, Continuous and Approximative Comparison

First use: S55 and S58.

Example Solved · Cooling

The model $T' = -0{,}1(T - 20)$, $T(0) = 80$ t in minutes, gives $T(t) = 20 + 60e^{-0{,}1t}$. With a step h, Euler gives $T_{n+1} = T_n - 0{,}1h(T_n - 20)$ in moments $t_n = nh$. For $h = 2$, on obtains 80,68,58,4. These are approximations of the continuous model at the moments 0.2,4; a law discrete physical given independently is not automatically Euler.

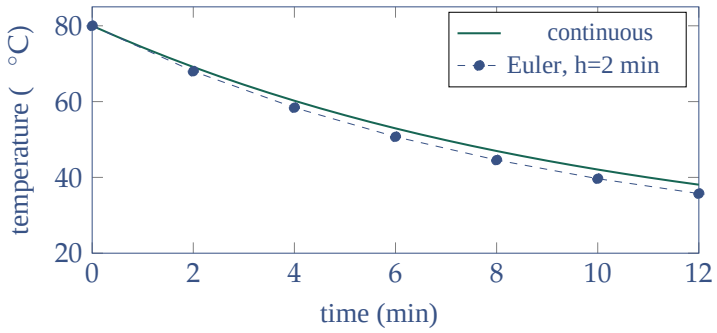


Figure 7.2. Read the deviation at t=6 min; segments connect successive approximations.

A6 · Choice of strategy Determine the first moment the continuous model reaches 50 degrees Celsius. Compare to the first value for Euler, with h=2 min, which is less than or equal to 50.
why the two results differ.

A first step: First solve an exponential equation; to Euler, write the formula of $T_n - 20$.

Wait a minute at the courthouse. Say what you mean by it, y' and t, then verify that the units of both members a differential equation is compatible.

Review postponed · S78. Complete the table of antiderivatives with trigonometric functions and control compositional factors.

Research activity

Cooling a drink · 40 min.

A drink initially at 80 °C is placed in a room held at 20 °C. For $t \geq 0$, note $T(t)$ its temperature in degrees Celsius, t the time elapsed in minutes. We admit the model $T'(t) = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial problem and give expression to $T(t)$.

2. Study the direction of variation in T and determine its limit in $+\infty$. Interpret results.

3. Determine the first time the temperature is 40 °C. Give the exact value, then a value approached in minutes.

4. Apply two steps of the Euler method with a step of 1 minute. Compare the approximation obtained after two minutes at the exact value $T(2)$.

5. Indicate a concrete situation in which the hypothesis of a constant ambient temperature would no longer be suitable.

Progressive exercises

E1 · Direct application. Determine all antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Application and justification. Determine the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ who verifies $F(0) = 0$.

E3 · Method to be organised. Determine the solution on $\mathbb{R}$ differential equation $y' = 3y$ who checks $y(1) = 2$.

E4 · Choice of strategy. Determine the solution on $\mathbb{R}$ differential equation $y' = -y + 4$ which checks $y(0) = 7$.

E5 · Synthesis or demonstration. The differential equation is considered $(E) : y' = y - x$ on $\mathbb{R}$. Verify that $g(x) = x + 1$ is a solution, then determine the solution of (E) who verifies $y(0) = 0$.

Work in the manual or annal

Barbazo, p. 228, Exercises 25 and 30. In S55: Exercise 30, to reveal a constant solution.
At home: Exercise 25 in full, distinguishing a condition on f a condition on f'.

Duty at home

DM7 · An evolution with variable input · 60 min.

We're looking for a function y defined and derivable on $[0; +\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Determine two actuals a and b so that $g(t) = at + b$ or a differential equation solution.

2. Deduce the expression of the solution that verifies $y(0) = 3$.

3. Study the variations of this solution and specify its minimum as well as the moment it is reached.

4. Write a Python argument function N, integer strictly positive, which applies the method d-Euler on $[0; 2]$ with the step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.

5. Why is the verification of differential equality in just a few moments not enough to prove that a function is a solution?

QCM A · antiderivative families

After S53; QCM B after S58.

For each question, only one proposal is accurate.
consult the correction.

1. A antiderivative of x^2 on $\mathbb{R}$ is:

A. $2x$ B. x^3

C. $x^3/3$ D. $3x^3$

2. All antiderivatives of e^{2x} are:

A. $e^{2x} + C$ B. $\frac{1}{2}e^{2x} + C$

C. $2e^{2x} + C$ D. $xe^{2x} + C$

3. A antiderivative of $2x/(1 + x^2)$ on $\mathbb{R}$ is:

A. $\frac{1}{1+x^2}$ B. $2\ln x$

C. $\ln x$ D. $\ln(1 + x^2)$

4. Two antiderivatives of the same function over an interval have curves:

A. Images of one another by vertical translation B. Necessarily identical

C. Always stinging D. Images by any horizontal translation

5. The antiderivative of $2x$ verifier $F(1) = 3$ is:

A. $x^2 + 3$ B. $2x^2 + 1$

C. $x^2 + 2$ D. $x^2 - 2$

QCM B · Differential Equation and Model

For each question, only one proposal is accurate.
consult the correction.

6. The solution of $y' = -2y$, $y(0) = 3$, is:

A. $3e^{2t}$ B. $3e^{-2t}$

C. $-2e^{3t}$ D. $3 - 2t$

7. The constant solution of $y' = -2y + 6$ is:

A. 6 B. -3

C. 2 D. 3

8. If y_p is a solution of $y' = ay + b(t)$, then $z = y - y_p$ verify:

A. $z' = az$ B. $z' = az + b(t)$

C. $z' = b(t)$ D. $z' = 0$ necessarily

9. Step by step h for $y' = -y$, from $y_0 = 2$, gives:

A. $y_1 = 2 + h$ B. $y_1 = e^{-h}$

C. $y_1 = 2 - 2h$ D. $y_1 = 2 - h^2$

10. If t is in minutes and T in degrees Celsius, the coefficient k of $T' = -k(T - 20)$ expressed in:

A. minutes B. Inverse minutes

C. degrees Celsius D. litres

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A antiderivative is unique without additional conditions.

2. Euler gives the exact value for any differential equation.

3. Checking a solution consists of checking the equation and the initial condition.